



Assignment number: 1

Complete actual name: Anoosha Irfan

Registration number: FA24B1-SE-048

Course code: SE-665

Section: A

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

Update the Ubuntu package index and install Docker Engine and curl.

```
anooshairfan@ubuntu:~$ sudo curl -SL https://github.com/docker/compose/releases/latest/download/docker-compose-linux-x86_64 -o /usr/local/lib/docker/cli-plugins/docker-compose
% Total % Received % Xferd Average Speed Time Time Time Current
         Dload Upload Total Spent Left Speed
  0     0    0     0     0      0     0     0     0  0:00  0:00  0:00     0
  0     0    0     0     0      0     0     0     0  0:00  0:00  0:00     0
100 31.14M 100 31.14M    0     0 17872      0  30:27  30:27  0:00 23684
anooshairfan@ubuntu:~$ sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
anooshairfan@ubuntu:~$ docker compose version
Docker Compose version v5.6.0
anooshairfan@ubuntu:~$ _
```

Create the system-wide Docker CLI plug-in directory and download the Docker Compose plug-in for an x86_64 Ubuntu server. Start the Docker service and add your current Ubuntu account to the docker group:

```
anooshairfan@ubuntu:~$ ssh anooshairfan@192.168.105.129
The authenticity of host '192.168.105.129 (192.168.105.129)' can't be established.
ED25519 key fingerprint is: SHA256:XBGE9epSK1uaYXksJJniC6v0x7MMqQLVbLkpIM1BkSE
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.105.129' (ED25519) to the list of known hosts.
anooshairfan@192.168.105.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Oct  5 07:38:53 AM UTC 2026

System load:  0.12               Processes:           230
Usage of /:   32.5% of 17.77GB   Users logged in:    0
Memory usage: 36%               IPv4 address for ens33: 192.168.105.129
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

35 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Sat Sep 12 17:32:14 2026 from 192.168.105.1
anooshairfan@ubuntu:~$
```

Verify the Git, Docker, and Docker Compose installations:

```
anooshairfan@ubuntu:~$ git --version
git version 2.53.0
anooshairfan@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
anooshairfan@ubuntu:~$ docker compose version
Docker Compose version v5.6.0
anooshairfan@ubuntu:~$ _
```

Clone the instructor-provided Gitea setup repository on the Ubuntu server:

```
anooshairfan@ubuntu:~$ cd ~
anooshairfan@ubuntu:~$ git clone https://github.com/WaqasSaleem97/Gitea.git
Cloning into 'Gitea'...
remote: Enumerating objects: 16, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 16 (delta 4), reused 7 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (16/16), 5.37 KiB | 343.00 KiB/s, done.
Resolving deltas: 100% (4/4), done.
anooshairfan@ubuntu:~$ cd Gitea
anooshairfan@ubuntu:~/Gitea$ _
```

Start Gitea and PostgreSQL:

```
anooshairfan@ubuntu:~/Gitea$ docker compose up -d
[+] up 25/25
  V Image gitea/gitea:latest Pulled
  V Image postgres:17-alpine Pulled
  V Network webnet Created
  V Volume gitea Created
  V Volume gitea_postgres Created
  V Container gitea Started
  V Container gitea_db Started
anooshairfan@ubuntu:~/Gitea$
```

Confirm that both containers are running and Confirm that Gitea responds locally on port 3000:

```
anooshairfan@ubuntu:~/Gitea$ docker compose ps
NAME      IMAGE              COMMAND                                SERVICE  CREATED          STATUS          PORTS
gitea     gitea/gitea:latest "/usr/bin/entrypoint..."           gitea    About a minute ago Up About a minute 22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db  postgres:17-alpine "docker-entrypoint.s..."           db       About a minute ago Up About a minute 5432/tcp
anooshairfan@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "%{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
```

On the initial Gitea installation page,

Initial Configuration

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).

Database Type *	PostgreSQL
Host *	db:5432
Username *	gitea
Password *	*****
Database Name *	gitea
SSL *	Disable
Schema	

Leave blank for database default ("public").

General Settings

Site Title *	Gitea: Git with a cup of tea
You can enter your company name here.	
Data Path *	/data/gitea
Most Gitea data will be saved to this directory.	

Run As Username	git
The operating system username that Gitea runs as, it must have write access to the data paths. This value is auto-detected and cannot be changed here. To use a different user, restart Gitea under that account.	

SSH Server Port	22
Port number your SSH server listens on. Leave empty to disable.	

HTTP Server Port *	3000
Port number the Gitea web server will listen on.	

Gitea Website URL *	http://192.168.105.129:3000/
Gitea web application's public URL for HTTP(S) access, clone URLs and email notifications.	

- ☐ Only administrators can create user accounts (no self registration)
- ☐ Enable registration CAPTCHA
- ☐ Require sign-in to view pages (recommended for private instances)
- ☒ Enable update checker

Optional Settings

- Email Settings
- Server and Third-Party Service Settings
- ▼ Administrator Account Settings

Creating an administrator account is optional. The first registered user will automatically become an administrator.

Administrator Username	anoochairfan
Email Address	anoochayirfan2006@gmail.com
Password	*****
Confirm Password	*****

These configuration options will be written into: `/data/gitea/conf/app.ini` 

To fine tune more details: edit the configuration file after installation. Reference: [app.example.ini](#).

Install Gitea

The screenshot must show your <github-username>@ubuntu prompt, the docker compose ps command,

```
anooshairfan@ubuntu:~/Gitea$  
anooshairfan@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/  
Gitea HTTP status: 200  
anooshairfan@ubuntu:~/Gitea$
```

Step 1.2: Create and Push the Initial Repository

The screenshot shows the 'New Repository' form in Gitea. The form is titled 'New Repository' and includes a sub-header: 'A repository contains all project files, including revision history. Already hosting one elsewhere? [Migrate repository.](#)'. The form fields are as follows:

- Owner:** A dropdown menu showing 'anooshairfan'.
- Repository Name:** A text input field containing 'Assignment01'.
- Visibility:** A checkbox labeled 'Make repository private' is unchecked. Below it, a note states: 'Only the owner or the organization members if they have rights, will be able to see it.'
- Description:** A text input field with the placeholder 'Enter short description (optional)'.
- Template:** A dropdown menu with the option 'Select a template.'
- Issue Labels:** A dropdown menu with the option 'Select an issue label set.'
- .gitignore:** A dropdown menu with the option 'Select .gitignore templates.' Below it, a note states: 'Choose which files not to track from a list of templates for common languages. Typical artifacts generated by each language's build tools are included on .gitignore by default.'
- License:** A dropdown menu with the option 'Select a license file.' Below it, a note states: 'A license governs what others can and can't do with your code. Not sure which one is right for your project? See [Choose a license.](#)'
- README:** A dropdown menu with the option 'Default'. Below it, a note states: 'This is the place where you can write a complete description for your project.'
- Initialize Repository:** A checkbox labeled 'Initialize Repository (Adds .gitignore, License and README)' is unchecked.
- Default Branch:** A text input field containing 'main'. Below it, a note states: 'The default branch is the base branch for pull requests and code commits.'
- Object Format:** A text input field containing 'sha1'. Below it, a note states: 'Object format of the repository. Cannot be changed later. SHA1 is most compatible.'
- Template:** A checkbox labeled 'Make repository a template' is unchecked.

At the bottom of the form is a blue button labeled 'Create Repository'.

create a separate local repository outside both the Gitea directory and the existing CC repository. Create a README.md containing:

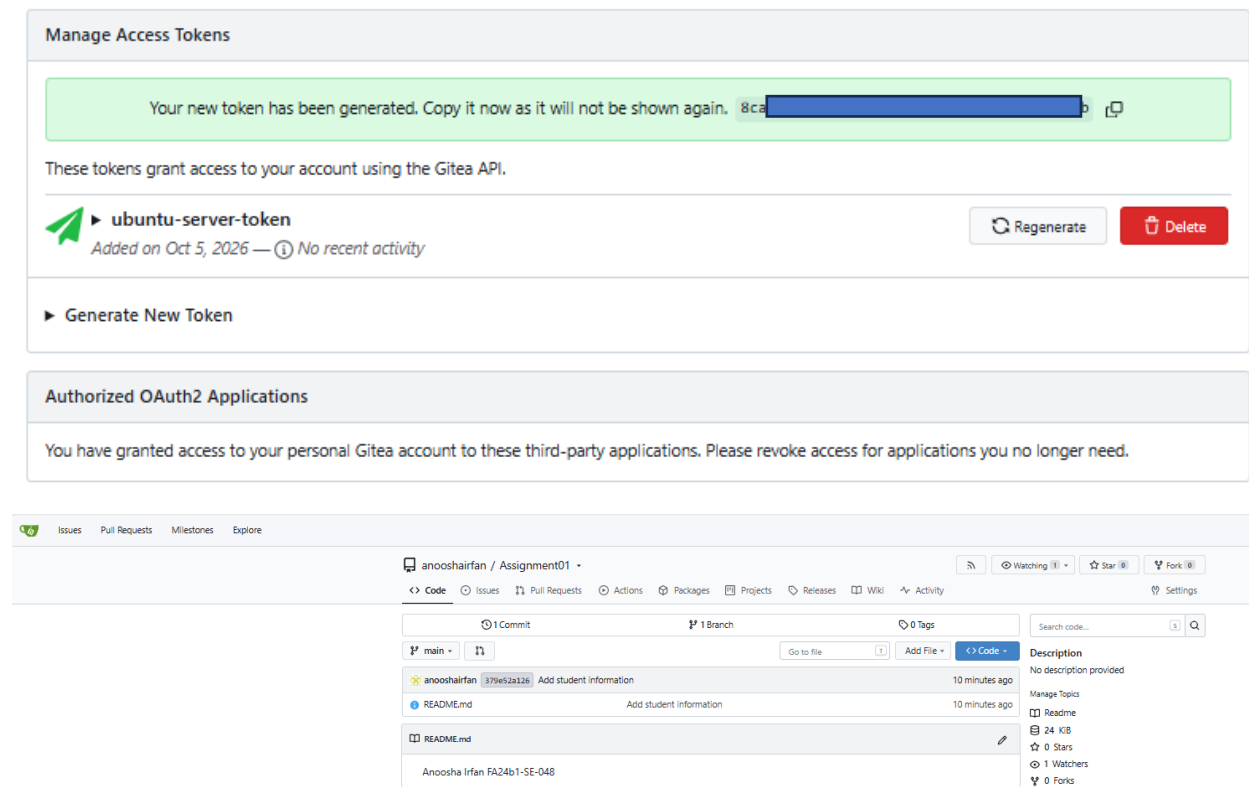
- your full name; and
- your registration number. Commit the README:

```

anooshairfan@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://anooshairfan@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 255 bytes | 255.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/anooshairfan/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
anooshairfan@ubuntu:~/Assignment01$ _

```

generate a personal access token:



The screenshot shows the Gitea user settings interface. The 'Manage Access Tokens' section displays a message: 'Your new token has been generated. Copy it now as it will not be shown again.' followed by a token starting with '8ca'. Below this, it states 'These tokens grant access to your account using the Gitea API.' and lists a token named 'ubuntu-server-token' added on Oct 5, 2026, with 'No recent activity'. There are 'Regenerate' and 'Delete' buttons for this token. A 'Generate New Token' link is also present. The 'Authorized OAuth2 Applications' section shows a message: 'You have granted access to your personal Gitea account to these third-party applications. Please revoke access for applications you no longer need.'

Below the settings, the repository page for 'anooshairfan / Assignment01' is visible. It shows 1 commit, 1 branch (main), and 0 tags. The commit history shows a commit by 'anooshairfan' (379e52a12e) titled 'Add student information' from 10 minutes ago. The file list shows 'README.md' (24 KiB, 0 stars, 1 watchers, 0 forks) with a description 'No description provided'.

add the local Gitea address as the gitea remote and Verify that the remote URL

```

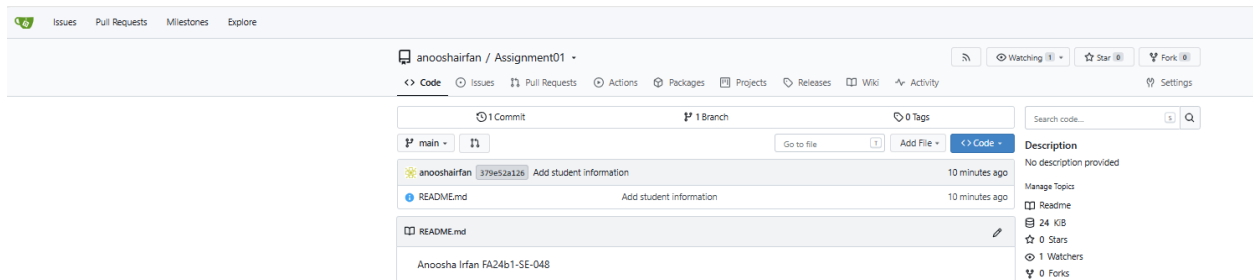
anooshairfan@ubuntu:~/Assignment01$ cd ~/Assignment01
anooshairfan@ubuntu:~/Assignment01$ git remote add github https://github.com/anooshairfan/Assignment01.git
anooshairfan@ubuntu:~/Assignment01$ git remote -v
gitea  http://anooshairfan@127.0.0.1:3000/anooshairfan/Assignment01.git (fetch)
gitea  http://anooshairfan@127.0.0.1:3000/anooshairfan/Assignment01.git (push)
github https://github.com/anooshairfan/Assignment01.git (fetch)
github https://github.com/anooshairfan/Assignment01.git (push)
anooshairfan@ubuntu:~/Assignment01$ _

```

Push the initial commit:

```
anooshairfan@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': anooshairfan
Password for 'https://anooshairfan@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 255 bytes | 42.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/anooshairfan/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
anooshairfan@ubuntu:~/Assignment01$ _
```

Step 1.3: Verify the Gitea Repository



Task 2: Push the Same Repository to GitHub

Step 2.1: Create the GitHub Repository and Verify Both Remotes

New repository

Q Type to search

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner *

anooshairfan

Repository name *

Assignment01

Assignment01 is available.

Great repository names are short and memorable. How about [cautious-umbrella](#)?

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

anooshairfan / Assignment01

Q T

Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

Assignment01

Public

Pin

Watch

main

1 Branch

0 Tags

Go to file

T

Add file

<> Code

anooshairfan

Add student information

379e52a · 35 minutes ago

1 Commit

README.md

Add student information

35 minutes ago

README

Anoosha Irfan FA24b1-SE-048

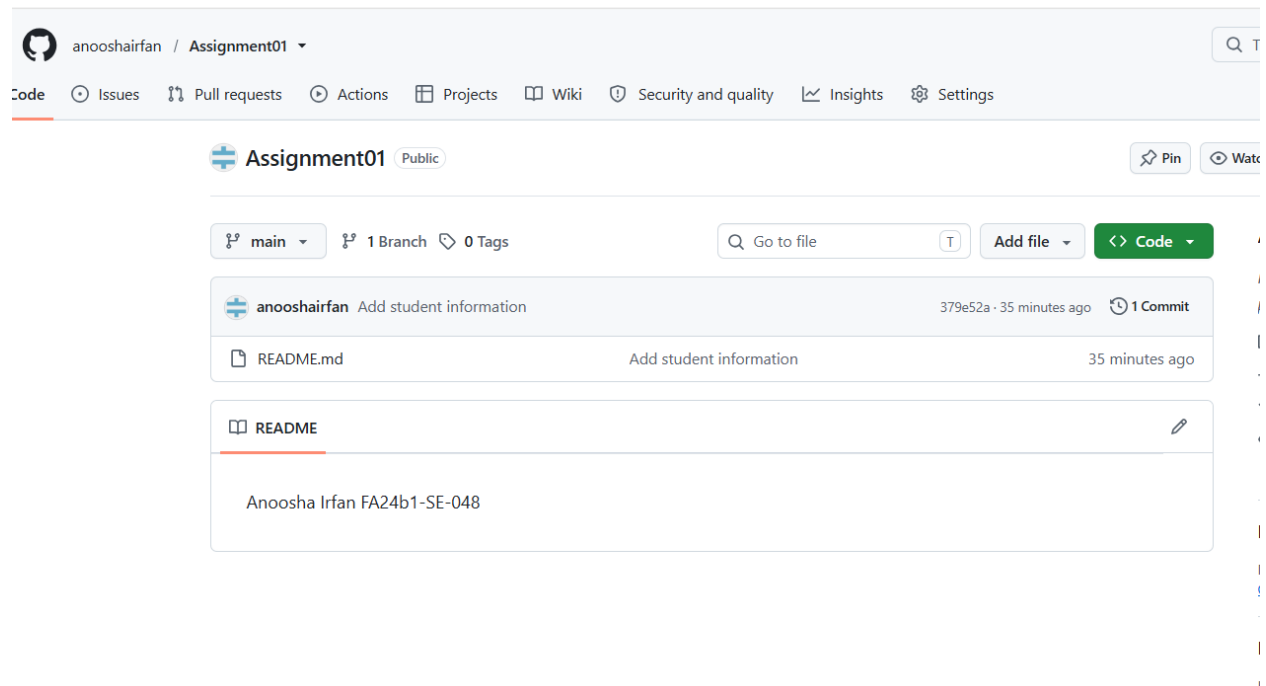
Add GitHub as the second remote and Verify both remotes:

```
anooshairfan@ubuntu:~/Assignment01$ cd ~/Assignment01
anooshairfan@ubuntu:~/Assignment01$ git remote add github https://github.com/anooshairfan/Assignment01.git
anooshairfan@ubuntu:~/Assignment01$ git remote -v
gitea    http://anooshairfan@127.0.0.1:3000/anooshairfan/Assignment01.git (fetch)
gitea    http://anooshairfan@127.0.0.1:3000/anooshairfan/Assignment01.git (push)
github   https://github.com/anooshairfan/Assignment01.git (fetch)
github   https://github.com/anooshairfan/Assignment01.git (push)
anooshairfan@ubuntu:~/Assignment01$ _
```

Step 2.2: Push to GitHub

```
anooshairfan@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': anooshairfan
Password for 'https://anooshairfan@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 255 bytes | 42.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/anooshairfan/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
anooshairfan@ubuntu:~/Assignment01$ _
```

Step 2.3: Verify the GitHub Repository



The screenshot shows the GitHub interface for a repository named 'Assignment01' owned by 'anooshairfan'. The repository is public. The main branch is selected, showing 1 branch and 0 tags. A commit titled 'Add student information' by 'anooshairfan' is displayed, dated 35 minutes ago. The commit message is 'Add student information'. Below the commit, the README file is shown with the content 'Anoosha Irfan FA24b1-SE-048'.

Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

```
cat .gitattributesanooshairfan@ubuntu:~/Assignment01$
anooshairfan@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
anooshairfan@ubuntu:~/Assignment01$
anooshairfan@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
anooshairfan@ubuntu:~/Assignment01$
anooshairfan@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
anooshairfan@ubuntu:~/Assignment01$
anooshairfan@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
anooshairfan@ubuntu:~/Assignment01$ _
```

Step 3.2: Create and Verify Three LFS Files

```
anooshairfan@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
anooshairfan@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
anooshairfan@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
anooshairfan@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
anooshairfan@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
anooshairfan@ubuntu:~/Assignment01$
```

Stage .gitattributes and the three files and Verify that all three files are tracked by Git LFS and display their sizes:

```
anooshairfan@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
anooshairfan@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
anooshairfan@ubuntu:~/Assignment01$
```

Step 3.3: Commit and Push the LFS Files

```
anooshairfan@ubuntu:~/Assignment01$ git commit -m "Add three files using Git LFS"
[main a0a747d] Add three files using Git LFS
4 files changed, 10 insertions(+)
create mode 100644 .gitattributes
create mode 100644 large-file-1.bin
create mode 100644 large-file-2.bin
create mode 100644 large-file-3.bin
anooshairfan@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': anooshairfan
Password for 'https://anooshairfan@github.com':
Username for 'https://github.com': anooshairfan
Password for 'https://anooshairfan@github.com':
Uploading LFS objects: 100% (3/3), 568 MB | 668 KB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 799 bytes | 99.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/anooshairfan/Assignment01.git
  379e52a..a0a747d  main -> main
anooshairfan@ubuntu:~/Assignment01$
```

Task 4: Create a Portfolio or CV with GitHub Pages

4.1: Create and Push the Portfolio Repository

The screenshot displays a GitHub repository interface for the user 'anooshaifan'. The repository is named 'anooshaifan.github.io'. The top navigation bar includes links for Code, Issues, Pull requests, Actions, Projects, Wiki, Security and quality, Insights, and Settings. The left sidebar shows the file structure with 'main' as the selected branch. The files listed are README.md, about.html, index.html, and style.css. The main content area shows the 'style.css' file, which contains CSS code for a portfolio layout. The code includes a 'body' selector with font-family, line-height, margin, padding, background-color, and color. It also includes a 'header' selector with text-align, background, color, padding, and border-radius. A 'section' selector is defined with background, margin, padding, border-radius, and box-shadow. An 'h2' selector is defined with color, border-bottom, and padding-bottom. A 'ul' selector is defined with list-style-type: square. The bottom section of the screenshot shows the 'index.html' file in the 'Edit' view. The code is a valid HTML5 document with a 'main' section containing a 'h1' and a 'h2'. It also includes a 'ul' with a single list item. The footer contains a 'p' with the text '© 2024 Anosha Tripathi'.

anooshaifan / anooshaifan.github.io

Code Issues Pull requests Actions Projects Wiki Security and quality Insights Settings

Files

main

Go to file

README.md

about.html

index.html

style.css

anooshaifan.github.io / style.css

anooshaifan Create style.css

Code Blame 36 lines (36 loc) · 635 Bytes

```
1 body {
2   font-family: Arial, sans-serif;
3   line-height: 1.6;
4   margin: 0;
5   padding: 20px;
6   background-color: #f4f4f9;
7   color: #333;
8 }
9 header {
10  text-align: center;
11  background: #0056b3;
12  color: white;
13  padding: 20px 0;
14  border-radius: 8px;
15 }
16 section {
17  background: white;
18  margin: 15px 0;
19  padding: 20px;
20  border-radius: 8px;
21  box-shadow: 0 2px 5px rgba(0,0,0,0.1);
22 }
23 h2 {
24  color: #0056b3;
25  border-bottom: 2px solid #0056b3;
26  padding-bottom: 5px;
27 }
28 ul {
29  list-style-type: square;
30 }
31 footer {
```

anooshaifan.github.io / index.html

Edit Preview

Cancel changes Commit changes

Spaces 4 No wrap

```
14 <section id="about">
15   <h2>About Me</h2>
16   <p>Welcome to my portfolio! I am a student working on cloud computing, DevOps, and version control tools like Git and GitHub.</p>
17 </section>
18
19 <section id="education">
20   <h2>Education</h2>
21   <p><strong>BSc Computer Science / IT</strong> - University / College</p>
22 </section>
23
24 <section id="skills">
25   <h2>Skills</h2>
26   <p><strong>Frontend</strong>: HTML5, CSS3, JavaScript, jQuery, Bootstrap</p>
27   <p><strong>Backend</strong>: Python, Django, Flask, Node.js, Express.js</p>
28   <p><strong>Cloud</strong>: AWS, Azure, Google Cloud Platform</p>
29   <p><strong>Tools</strong>: Git, Docker, Kubernetes, Jenkins, Terraform</p>
30   <p><strong>Languages</strong>: Python, JavaScript, Java, C++, C#</p>
31 </section>
32
33 <section id="projects">
34   <h2>Projects</h2>
35   <p><strong>Project 1</strong>: A web application for managing tasks and projects using React and Node.js.</p>
36   <p><strong>Project 2</strong>: A REST API for a blog using Django and PostgreSQL.</p>
37   <p><strong>Project 3</strong>: A cloud-based file storage system using AWS S3 and Python.</p>
38   <p><strong>Project 4</strong>: A microservices architecture for an e-commerce application using Docker and Kubernetes.</p>
39   <p><strong>Project 5</strong>: A data analytics dashboard using Python, Django, and Chart.js.</p>
40 </section>
41
42 <section id="contact">
43   <h2>Contact</h2>
44   <p>Feel free to reach out to me at <a href="mailto:anoosha@anoosha.com">anoosha@anoosha.com</a> or <a href="https://www.linkedin.com/in/anoosha-tri">LinkedIn</a>.</p>
45 </section>
46 </html>
```

Use Control + Shift + E to toggle the _tab_ key moving focus. Alternatively, use _esc_ then _tab_ to move to the next interactive element on the page.

Step 4.2: Publish with GitHub Pages

GitHub Pages

GitHub Pages is designed to host your personal, organization, or project pages from a GitHub repository.

Your site is live at <https://anooshairfan.github.io/>
Last [deployed](#) by  [anooshairfan](#) 3 minutes ago

 Visit site

Unpublish site

Build and deployment

Source

Deploy from a branch ▾

Branch

Your GitHub Pages site is currently being built from the `main` branch. [Learn more about configuring the publishing source for your site.](#)



main ▾



/ (root) ▾

Save

Learn how to [add a Jekyll theme](#) to your site.

Your site was last deployed to the [github-pages](#) environment by the [pages build and deployment](#) workflow.
[Learn more about deploying to GitHub Pages using custom workflows](#)

Custom domain

Step 4.3: Verify the Live Portfolio

← ↻  <https://anooshairfan.github.io/index.html>

Anoosha Irfan

Student / Developer Portfolio

About Me

Welcome to my portfolio! I am a student working on cloud computing, DevOps, and version control tools like Git and GitHub.

Education

- BS SE - University / College

Skills

- Git & GitHub / Gitea
- Docker & Linux Terminal
- HTML5 & CSS3

Projects

- **Assignment 01:** Multi-Task Lab with Docker, Gitea, Git LFS, and GitHub Pages.

[About me](#)

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